

Correctness-Gated Policy Optimization for Streaming-Accelerator RTL: A Failure-Aware Post-Route Study

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Abstract—Improving the best generated register-transfer-level (RTL) implementation does not establish that a generator produces useful hardware more reliably. We study correctness-gated policy optimization for streaming accelerators using an endpoint that preserves sample multiplicity and assigns zero frequency to incorrect or implementation-failed draws. An executable oracle gates a learned physical reward during training; Vivado post-route implementation supplies the evaluation evidence. On a separately frozen 20-design split, two policy-training seeds sharing one supervised fine-tuned (SFT) checkpoint and a canonicalization-aware random-forest (RF) reward raise failure-penalized timing-derived maximum frequency from 35.4 to 93.0 MHz. The paired gain is 57.6 MHz, with a 95% design-bootstrap interval of [37.7, 77.2] MHz. Both RF seeds improve; a single-seed correctness-only control improves functional yield but lowers the physical endpoint. RF also exceeds the original multilayer-perceptron reward package, without isolating model class from representation changes. The gain comes with more LUTs and registers, fewer DSP blocks, slightly higher vectorless power, and a small matched VerilogEval regression. Post-primary setup-closure and board checks support the physical direction on descriptive subsets, while median-filter results remain weak. These findings support a bounded distribution-level improvement within the studied accelerator generators, not general RTL superiority or a universal power–performance–area gain.

Index Terms—RTL generation, reinforcement learning, large language models, FPGA physical design, hardware-aware optimization, reward-model reliability

I. INTRODUCTION

Executable RTL benchmarks have made functional correctness measurable for language-model-generated hardware [1], [2]. Physical usefulness is a different capability. Behaviorally compatible modules can implement the same computation with sharply different critical paths, resource mappings, and achievable clocks. A model trained only to pass a testbench has no reason to prefer the physically better form. Conversely, generating many candidates and selecting the fastest after electronic-design-automation (EDA) evaluation shifts the cost to inference and repeated physical design.

Recent work has therefore connected RTL generation to synthesis feedback, reinforcement learning, evolutionary search, or per-design adaptation [3]–[9]. A central measurement problem remains: an optimizer can improve a selected candidate, a conditional-on-correct mean, or the learned objective without making a random policy draw more useful. Physical implementation is also too expensive for every policy update,

which makes a learned reward attractive but exposes it to distribution shift and overoptimization [10]. The reward is least trustworthy where the optimized policy places its mass.

The open question in this paper is therefore deliberately bounded: within a controlled streaming-accelerator domain, does proxy-guided training change the generator’s physical utility at a fixed draw budget after functional and implementation failures are charged to the policy that produced them? Answering that question also requires distinguishing a timing-aware reward from generic policy movement and from an increase in correctness alone. The group- relative update rule and EDA feedback are ingredients of the study rather than new algorithms; the contribution is the failure-aware distribution-level formulation, controlled reward-package comparisons, and a prospective post-route test.

We address these two issues with a strict training/evidence boundary (Fig. 1). During training, an executable oracle first rejects functionally invalid RTL; a learned physical reward ranks only admissible completions. During reporting, neither the reward nor a selected example is accepted as hardware evidence. Every distinct correct candidate is implemented by the same post-route Vivado flow, its original sample multiplicity is restored, and every incorrect or implementation-failed draw is assigned zero frequency. The resulting equal-sample endpoint measures the physical utility of one draw while keeping the design, rather than the candidate, as the unit of aggregation.

The primary test is a separately frozen 20-design study spanning interpolation and extrapolation across the retained accelerator generators. Two policy-training seeds sharing one SFT checkpoint and a canonicalization-aware random-forest (RF) reward raise penalized equal-sample F_{max} from 35.4 to 93.0 MHz. Their seed-averaged paired effect is 57.6 MHz with interval [37.7, 77.2] MHz; each training seed is positive, and the interpolation and extrapolation intervals both exclude zero. The optimized policy’s correctness remains close to the supervised fine-tuned (SFT) baseline, and the failure penalty already charges the endpoint for the difference.

Matched controls constrain the explanation for the gain. The repaired RF reward exceeds the original multilayer-perceptron (MLP) reward by 26.9 MHz under the same tasks and sampling budget. Correctness-only training instead raises validity to 67.4% but lowers penalized F_{max} relative to SFT in its single preregistered seed. The observed physical gain therefore does not arise from higher functional yield; attribution beyond these realized checkpoints requires further training replication. Conditional resource and power analysis exposes its imple-

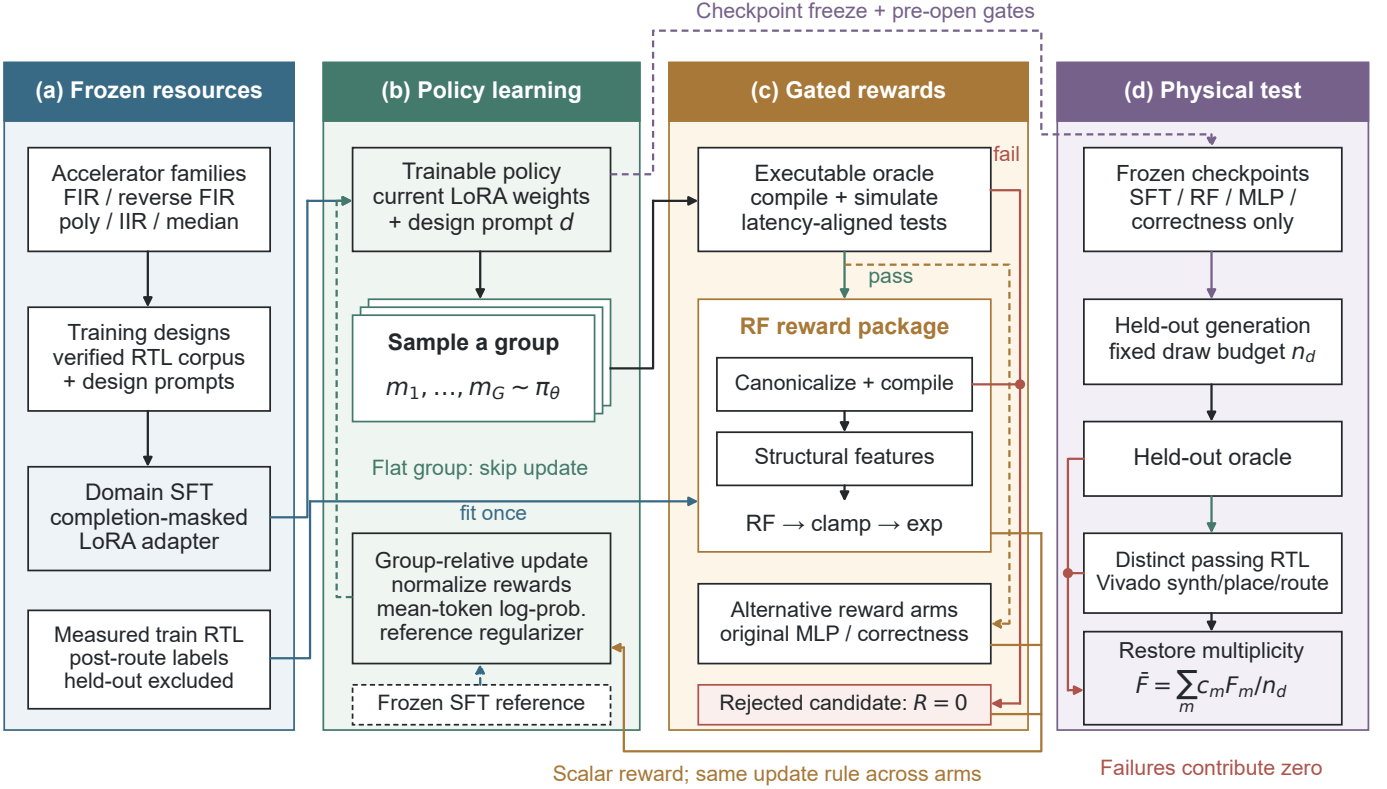


Fig. 1. Implemented training and physical-evaluation workflow. (a) Domain SFT and the physical predictor use training-only resources. (b) Fresh policy samples receive group-relative updates with a frozen SFT reference; flat groups do not advance the update counter. (c) The RF reward uses canonicalized structural features; MLP and correctness-only rewards are separate control runs. Red branches assign zero to rejected candidates. (d) Frozen checkpoints are evaluated using the executable oracle and post-route implementation. Original multiplicity and the full draw denominator are restored; incorrect and implementation-failed draws are not discarded. Dashed paths denote feedback, references, or separately run controls.

mentation cost. A separate, explicitly descriptive PYNQ-Z2 sweep is retained in the evidence record rather than used to strengthen the primary population claim.

This paper makes four contributions:

- *Controlled empirical finding:* on the frozen accelerator split, both independently trained RF policies improve the failure-penalized post-route endpoint under interpolation and extrapolation. Descriptive family and sensitivity analyses expose, rather than average away, the weak family.
- *Reward-package comparison:* original-reward and correctness-only arms separate the observed gain from an increase in functional yield. The comparison supports the repaired reward package on the studied split; it does not isolate random-forest architecture from canonical features.
- *Failure-aware protocol:* the fixed-draw estimand preserves original sample multiplicity and charges every incorrect or implementation-failed draw zero, while real post-route EDA rather than the learned proxy remains the reporting authority.
- *Bounded engineering interpretation:* candidate probabilities, a source-level RTL contrast, conditional PPA, setup-closure and board sensitivities, and matched general-RTL evaluation expose both physical improvements and specialization costs.

The dominant observed mechanism is probability realloca-

tion, not demonstrated open-ended circuit invention. Supervision exposes multiple correct implementations, and policy optimization often makes fast forms more probable when the reward can distinguish them. The sealed support diagnostic also has exceptions, so the paper does not reduce every gain to reweighting within the finite SFT sample. This narrower interpretation predicts stronger gains where fast structures are representable and reward-distinguishable, weaker families, and failure when a learned proxy ceases to track physical reality.

II. RELATED WORK

A. Executable RTL generation

VerilogEval and RTLLM established simulation-backed evaluation for natural-language-to-RTL generation [1], [2]. VeriGen, RTLCode, CodeV, and CraftRTL improve generation through specialized pretraining, supervised adaptation, data construction, or targeted repair [11]–[14]. VeriSeek introduces code-structure-guided reinforcement learning, while VeriReason combines supervised learning, testbench feedback, and group-relative policy optimization [15], [16]. Formal-RTL couples executable specifications to formal verification in a multi-agent synthesis flow [17]. These systems make correctness an executable outcome. Our oracle serves the same indispensable role, but our dependent variable is the physical quality of the entire sampled policy distribution after correctness has been enforced.

B. LLM-driven physical optimization

Physical feedback enters current RTL workflows at several different loci. RTLRewriter rewrites an existing implementation [5]. PPA-RTL constructs synthesis-scored preference pairs and applies direct preference optimization (DPO) [3]. ChipSeek uses curriculum-guided dynamic policy optimization with syntax, functional, and synthesis-derived feedback [4]. Alpha-RTL instead adapts the policy for each target at test time through an executable EDA loop [8]. Dr. RTL uses commercial-tool feedback, critical-path analysis, parallel rewriting, and reusable optimization skills [9]. POET and Evolve apply evolutionary search to verified RTL variants [6], [7]. RTL-OPT contributes a benchmark of suboptimal/human-optimized RTL pairs and an automated correctness/PPA evaluation flow [18].

These approaches answer related but non-identical questions. Per-design adaptation, evolutionary search, and rewriting naturally evaluate the selected implementation returned by the search. Training before deployment asks whether the generator itself has improved before any post-generation selection. Our primary object is therefore the equal-draw empirical policy distribution: sample multiplicity, incorrect outputs, and implementation failures all remain in the denominator. The perfect-selector comparison separately gives inference-time search the most favorable possible oracle and asks how many SFT draws are needed to match one optimized draw. Table I makes these distinct optimization and reporting objects explicit. This is not a claim that previous studies report only their best sample: ChipSeek also reports mean and extreme physical metrics among correct generations. The distinction here is retaining the entire draw denominator, including failures, and testing the resulting endpoint with a frozen paired post-route protocol. The present controls are internal reward comparisons, not a reproduced head-to-head evaluation against PPA-RTL or ChipSeek.

C. Physical predictors and reward reliability

Pre-synthesis performance predictors reduce the cost of exploring FPGA design spaces [19]–[21]. They are useful training signals precisely because detailed implementation is costly, but optimizing a learned reward changes its input distribution. Reward-model overoptimization can magnify errors that appear benign on the fitting distribution [10]. We therefore separate the proxy’s role from the reporting authority: the repaired RF and original MLP rewards may update the policy, whereas only executable correctness and real post-route or board measurements support physical claims. The historical trajectory is used as a bounded diagnostic of this distinction, not as the paper’s headline or as evidence that every physical predictor fails.

D. Scope of the contribution

The novelty claimed here is not the invention of synthesis-aware reinforcement learning or of the group-relative policy update. It is a controlled way to ask whether training improves the generator itself: the primary object is the fixed-draw policy distribution, every functional and implementation failure

is retained, real post-route EDA is the reporting authority, and matched arms compare physical-reward packages with correctness pressure. The paper does not attempt free-form industrial RTL optimization, formal equivalence, or universally superior silicon. Within its controlled accelerator domain, this scope enables paired design-level analysis, candidate-level mechanism tests, and fail-closed claim traceability that are not interchangeable with reporting the best implementation returned by a search.

III. METHOD

A. Problem formulation

For a design specification d , let policy p sample an RTL completion m . The desired quantity is not the best implementation found after an unbounded search, but the expected post-route frequency of a policy draw when invalid outcomes remain possible. Training must therefore solve two coupled problems: preserve the specified input/output behavior and change the distribution over physically distinct correct implementations. Figure 1 separates the inexpensive feedback used for policy updates from the more expensive evidence used for reporting.

B. Executable functional gate

Each accelerator family has a deterministic software reference mapping an input stream to its required output stream. A generated module is compiled and simulated on frozen stimuli. Because admissible implementations may introduce different pipeline depths, the oracle searches a bounded latency window and accepts the best aligned match fraction at or above 0.999. The search allows zero through 40 cycles of latency and discards 8 warm-up outputs. Thus the verdict is binary, but the implemented threshold is not an exact-equality contract: a sufficiently long trace can pass despite a mismatch. We use “correct” below to mean oracle-passing, not formally equivalent. The same oracle implementation filters supervised examples, gates policy rewards, and evaluates held-out samples, with held-out stimulus streams reserved for evaluation.

Let $C_d(m)$ denote the binary oracle verdict. Every reward arm obeys

$$g_d(m) = \begin{cases} 1, & C_d(m) \text{ passes,} \\ 0, & C_d(m) \text{ fails.} \end{cases} \quad (1)$$

The gate rejects behavior changes detected by its sampled tests. It neither proves equivalence on all inputs nor establishes that a proxy ranks admissible modules properly; physical implementation addresses the latter question.

C. Supervised warm start and policy update

RTLCoder [12] is adapted on an oracle-verified accelerator corpus using completion-masked LoRA [22]. Held-out designs are excluded from the corpus, reward-training rows, and policy prompts. The supervised adapter supplies both the policy initialization and the frozen reference. Anchoring to the SFT

TABLE I

PROTOCOL-LEVEL POSITIONING AGAINST THE CLOSEST PHYSICAL RTL LEARNING AND OPTIMIZATION SETTINGS. “SELECTED IMPLEMENTATION” DENOTES THE OUTPUT OF A TARGET-SPECIFIC SEARCH; “POLICY DISTRIBUTION” RETAINS EVERY SAMPLED OUTCOME AND ITS MULTIPLICITY.

Approach	Optimization locus	Feedback/reporting authority	Primary object
PPA-RTL [3]	Fixed-preference DPO	Synthesis-scored preference pairs	Objective-conditioned RTL generation
ChipSeek [4]	Curriculum-guided policy optimization	Syntax, simulation, and synthesis feedback	Correctness and conditional physical quality
Alpha-RTL [8]	Per-design test-time learning	Executable verification and EDA feedback	Selected implementation after adaptation
Dr. RTL [9]	Agentic per-design rewriting	Commercial EDA and critical-path feedback	Selected optimized implementation
RTL-OPT [18]	Evaluation benchmark	Automated equivalence and PPA evaluation	Suboptimal-to-reference optimization
This work	Proxy-guided on-policy training	Executable training gate; post-route EDA for reporting	Equal-sample policy distribution on a frozen split

adapter rather than the raw base model avoids pulling the optimized policy away from the domain competence established by supervision.

Each training group contains 8 completions sampled at temperature 1.0, with at most 1536 completion tokens. If R_i is the gated reward of completion m_i , its within-group advantage is

$$A_i = \frac{R_i - \bar{R}}{s_R + \epsilon}. \quad (2)$$

Fresh completions are sampled from the current policy at each attempted group; this is on-policy, proxy-guided optimization, not offline reinforcement learning on a fixed transition or preference dataset. Groups without reward variation contain no ranking information and do not advance the optimizer-update counter. For an informative group, the implemented objective uses the mean completion-token log probability $\ell_\theta(m \mid d) = |m|^{-1} \sum_t \log \pi_\theta(m_t \mid d, m_{<t})$:

$$\mathcal{L}_i = -A_i \ell_\theta(m_i \mid d) + \beta (\exp(\Delta_i) - \Delta_i - 1), \quad (3)$$

where $\Delta_i = \ell_{\text{SFT}}(m_i \mid d) - \ell_\theta(m_i \mid d)$ and the reference is detached. Losses are averaged within the group. This length-normalized update is inspired by group-relative policy optimization [23]; it is not the standard clipped GRPO objective. Exponentiating a difference of mean token log probabilities does not yield the usual sequence-probability ratio, so the penalty is a reference regularizer, not an asserted unbiased KL estimate. Every arm uses AdamW with learning rate 1e-5, reference-penalty coefficient 0.1, and a target of 276 non-flat optimizer updates. Checkpoints are indexed by optimizer updates, not attempted groups; the frozen midpoint is update 138.

D. Physical reward and matched controls

The repaired reward first maps admissible RTL through a frozen lexical canonicalizer κ , then predicts log-frequency from canonical structural features ϕ :

$$R_d^{\text{RF}}(m) = g_d(m) \exp \left[\text{clip} \left(\widehat{f}_{\text{RF}}(\phi(\kappa(m))), a, b \right) \right]. \quad (4)$$

Here $\widehat{f}_{\text{RF}}$ predicts log-frequency, a, b are the frozen log-frequency clamp bounds, and exponentiation returns the reward in MHz. The RF contains 300 trees and uses 15 features: multiplier count, maximum per-statement multiplier count, positive-edge count, nonblocking and blocking assignment counts, multiply-accumulate structure, register and wire counts, ternary and comparison counts, statement count, a structural pipeline ratio, and loop count, maximum bound, and summed bounds. Descriptive impurity importance ranks multiplier count, multiply-accumulate structure, pipeline ratio, register count, and nonblocking assignments highest; this ranking describes the fitted predictor and is not a causal circuit attribution. It is fitted to 229 measured RTL rows that exclude every held-out and sealed design. Layout-dependent line counts and name-dependent features are absent. The canonicalizer is lexical and layout-invariant under its frozen mutation suite; it is not a canonical hardware representation. Canonicalization or canonical-compilation failure also sets the reward to zero and is logged.

This architecture is a disclosed post-hoc repair hypothesis derived from an older reward failure. Its feature representation and hyperparameters were fixed before the sealed efficacy split was opened; the sealed study is thus a prospective efficacy test of a post-hoc hypothesis, not a claim that the repair hypothesis itself was selected blindly.

Two controls compare reward packages. The original-MLP arm restores the previous textual/structural predictor while keeping the SFT initialization, prompts, oracle, optimizer, SFT reference, update target, and evaluation fixed. The correctness-only arm uses $R_d^{\text{C}}(m) = g_d(m)$ and therefore tests whether correctness pressure suffices in the realized training run. Because canonicalization, features, and predictor class change jointly, RF versus MLP is not a component-wise ablation. The RF arm has 2 independent training seeds, the MLP arm has 2, and the correctness-only arm has 1 preregistered full-budget seed.

The repaired reward is not operationally flat. It resolves 114 of 147 eligible RF groups, or 77.6%. Before sealed generation, all 801 distinct reward-eligible training candidates reproduced their logged canonical hash and features, compiled, and passed sampled original/canonical trace comparison; there were 0 rejections and 0 cross-design collisions. The trace check uses

256 vectors per frozen oracle seed and is evidence of the executed reward path, not formal equivalence.

E. Failure-penalized physical endpoint

After generation, each distinct oracle-correct candidate is implemented once and joined back to its original multiplicity. For policy p and design d , let $\mathcal{M}_{p,d}$ be the distinct correct candidates, c_m the sampled multiplicity, n_d the complete draw budget, and $F(m)$ the post-route timing-derived frequency, defined as zero when implementation fails. The primary per-design endpoint is

$$\bar{F}_{p,d} = \frac{\sum_{m \in \mathcal{M}_{p,d}} c_m F(m)}{n_d}. \quad (5)$$

Incorrect samples are absent from $\mathcal{M}_{p,d}$ but remain in n_d and therefore contribute zero. Each arm receives 48 draws per design; a two-seed arm uses 24 draws from each seed. The headline statistic is the unweighted mean of $\bar{F}_{p,d}$ across designs, so the design rather than the number of distinct candidates is the aggregation unit. Each candidate is implemented once under the same frozen clock request; $F(m)$ is therefore derived from that request and worst negative slack rather than from candidate-specific repeated timing closure. This symmetry supports the paired comparison but bounds absolute-frequency claims to the frozen flow, device, and request.

The endpoint decomposes as

$$\bar{F}_{p,d} = q_{p,d} \mu_{p,d}, \quad (6)$$

where $q_{p,d}$ is empirical correctness and $\mu_{p,d}$ is multiplicity-weighted frequency conditional on an oracle-passing draw, retaining zero for implementation failure. We set $\mu_{p,d} = 0$ when no draw passes. A policy that rarely emits very fast correct RTL can have high $\mu_{p,d}$ and low $\bar{F}_{p,d}$; reporting only the former would change the estimand.

For paired uncertainty, optimized-minus-SFT differences are computed per design and resampled within frozen family-by-regime strata. The primary plan uses 100,000 resamples and a 95% percentile interval. The direct RF-minus-MLP analysis reuses the identical resample plans but is labeled post-primary. Candidate draws are not treated as independent experimental units, and these design-level intervals do not estimate variation from SFT training, reward-data construction, or model-family choice.

F. Search and distribution diagnostics

For a perfect-selector comparison, the empirical SFT distribution contains every measured candidate with its multiplicity plus a zero outcome for every failure. If $G_d(x)$ is its empirical cumulative distribution, the maximum of N independent draws has cumulative distribution $G_d(x)^N$. This exact curve assumes access to every draw's measured timing-derived frequency before selection. It upper-bounds reranking on this finite sampled support, not on the unknown full SFT distribution. Draw count is not wall-clock cost: oracle calls,

uncached correct-draw implementation attempts, and content-deduplicated implementations are reported separately. Training cost is not amortized into these inference-only counts.

For mechanism, define the all-sample candidate mass $w_{p,d,m} = c_m/n_d$ and the dominant mass $\max_m w_{p,d,m}$. Normalizing over correct candidates gives $\tilde{w}_{p,d,m} = c_m/\sum_j c_j$ and effective support

$$D_{p,d} = \left(\sum_m \tilde{w}_{p,d,m}^2 \right)^{-1}. \quad (7)$$

Lower $D_{p,d}$ means probability is concentrated on fewer correct strings; it does not itself imply better hardware. We therefore pair it with real candidate frequencies and test whether SFT sampling already contains an implementation at least as fast as the optimized conditional mean.

G. Fail-closed result generation

The analysis requires the complete frozen design set, every declared arm and seed, the full sample denominator, and a physical-design row for every distinct correct candidate. Missing implementation rows are errors, not silently discarded observations. Separate generators emit named TeX claims, tables, figures, JSON results, input/output hashes, and file:line provenance. The verifier regenerates these outputs, resolves every claim used by the manuscript, and rejects empirical numbers typed directly into authored prose.

IV. EXPERIMENTAL PROTOCOL

A. Accelerator domain and sealed split

The benchmark contains streaming finite-impulse-response (FIR), reverse-FIR, polynomial, recursive-filter, and median-filter accelerators under a uniform clocked interface. The families cover feed-forward arithmetic pipelines, recurrences, and comparator networks. Each specification and its executable reference are generated programmatically, and several implementation styles provide physically distinct but behaviorally equivalent training examples. The domain is deliberately controlled: it spans multiple accelerator structures but not arbitrary industrial RTL.

The sealed evaluation contains 20 designs, balanced between 10 interpolation and 10 extrapolation tasks. Family quotas are unequal because the eligible median interpolation pool was empty; the median family is represented only under extrapolation, so median-family and regime effects cannot be separated. Table IV reports the exact composition. Interpolation changes parameters within training ranges; extrapolation uses larger or newly parameterized instances outside those ranges. Every sealed design is excluded from the SFT corpus, reward rows, policy prompts, and prior analysis. The split remained unopened until all declared checkpoints and pre-open validation artifacts were frozen.

B. Arms, seeds, and sampling

The primary comparison uses RTLCode, its completion-masked SFT adapter, and two independently trained repaired-RF policy adapters. The MLP arm trains two matched policies

from the same SFT adapter using the original reward. The correctness-only arm uses one preregistered full-budget seed. Arms are matched on non-flat optimizer updates rather than attempted groups, because a flat group produces no policy-ranking information. A crashed run is discarded and restarted from the frozen SFT state; partial histories are never merged. Equal non-flat updates do not imply equal training compute: flat groups still require generation and oracle execution, particularly for correctness-only rewards. The primary comparison matches deployment draws, not end-to-end training time or energy.

At the sealed endpoint, SFT and every trained arm receive 48 draws per design at temperature 1.0. Two-seed arms contribute 24 draws per seed. The midpoint diagnostic uses 8 draws per RF seed and design. Generation seeds are distinct from training seeds and from the 2 oracle streams, each of which contains 1,024 test vectors. Accepted candidates are deduplicated only for implementation; multiplicity is preserved for every policy statistic.

C. Post-route implementation and statistical analysis

The frozen physical flow is 2023.1 (laptop, xc7z020c1g400-1). Every distinct correct candidate is synthesized, placed, and routed with the same 5.0-ns requested clock. Achievable frequency is derived from the request and reported worst negative slack. We therefore use “post-route timing-derived F_{max} ” rather than “timing closure”: the flow does not repeatedly rerun implementation at candidate-specific clock periods. The historical script introduces the clock constraint after synthesis; this flow is held fixed rather than retroactively corrected in the sealed result.

The primary estimand is Eq. (5). Correctness and conditional frequency are diagnostics, not replacement endpoints. The primary RF-minus-SFT confidence plan is frozen and stratified by family and regime, with percentile bounds. RF-minus-MLP intervals reuse its exact resample plans as a bounded post-primary comparison. The family cells are small and unequal, so the manuscript adds an unstratified paired-bootstrap and leave-one-family-out sensitivity but does not replace the frozen primary plan. No multiplicity adjustment is applied: the RF-versus-SFT contrast is the sole prespecified primary inference, while control, family, PPA, and mechanism analyses are secondary or descriptive. Resource and power analyses condition on designs with compiled correct support in SFT, MLP, and RF; assigning zero area or power to a failed sample would incorrectly make failure look efficient. Power is Vivado’s vectorless estimate, not measured board power.

D. Provenance-bound PYNQ-Z2 protocol

Figure 2 shows the candidate-selection contract, implemented test harness, and trace-validation path.

The board study uses within-design SFT/RF pairs and applies the same multiplicity-aware candidate rule to both policies. All selected DUTs, a co-resident echo canary, and the selector interface occupy one bitstream/HWH pair. Each complete run sweeps the same clock grid, recomputes the

passing boundary from raw scores, and records the bitstream, HWH, selector-map, and selection-manifest hashes.

The first outcome-independent family hash selected a reverse-FIR design with no correct candidate in either paired arm, so that attempt stopped without a silicon outcome. Before any board measurement, eligibility was amended to require a correct candidate in both arms and the same outcome-independent hash was reapplied. The common sweep floor was also lowered after sealed Vivado rows showed that the original range would not bracket every selected DUT. No reward, timing, area, power, or resource outcome was used to choose the replacement. Nevertheless, these feasibility amendments make the resulting board subset descriptive rather than independently confirmatory.

E. Supporting and contextual evaluations

An earlier 30-design RTLCode evaluation provides broader five-family evidence and the candidate-distribution analysis. A separate Qwen-Coder experiment [24] repeats the full SFT-to-policy-optimization recipe on 22 designs from the earlier 3-family universe; it is not pooled with the sealed study. The post-primary perfect-selector analysis uses the sealed SFT support and retains the earlier evaluation only as historical context.

Three contextual controls answer narrower questions. A stored API experiment uses the same tasks and oracle with 8 samples per prompt arm. The HLS comparison measures post-implementation throughput for no-pragma and engineer-pragma C++ but compares them with the fastest measured optimized RTL candidate, not a policy average. The matched post-primary VerilogEval extension evaluates the sealed SFT and RF checkpoints using 20 samples per checkpoint on 156 problems. The earlier base/SFT/policy run uses a different generation budget and is not pooled with this comparison. Different budgets and selection rules prevent the API and HLS results from serving as causal primary baselines.

A completed post-primary pilot reimplements 10 selected SFT/RF candidates with candidate-specific period bracketing under Vivado 2026.1. Constraints precede synthesis in that flow. It therefore tests sensitivity to both constraint handling and tool version, not the timing estimator alone. Its pass criterion verifies setup slack, constraint coverage, and routing, but does not enforce hold or pulse-width closure. We report it as a descriptive setup-closure sensitivity and do not replace the primary population endpoint with this selected subset.

F. Study status and reproducibility

An earlier validation attempt used a stricter contract over every nonempty RF training completion. It stopped before sealed efficacy when that contract encountered a malformed, oracle-rejected completion. After diagnosing the scope mismatch, the present study froze a reward-eligible contract over exactly the candidates that could influence the RF reward, reused the already-trained checkpoints without retraining or selection, and opened a still-unseen sealed efficacy split only after the revised gate passed. The efficacy test is prospective

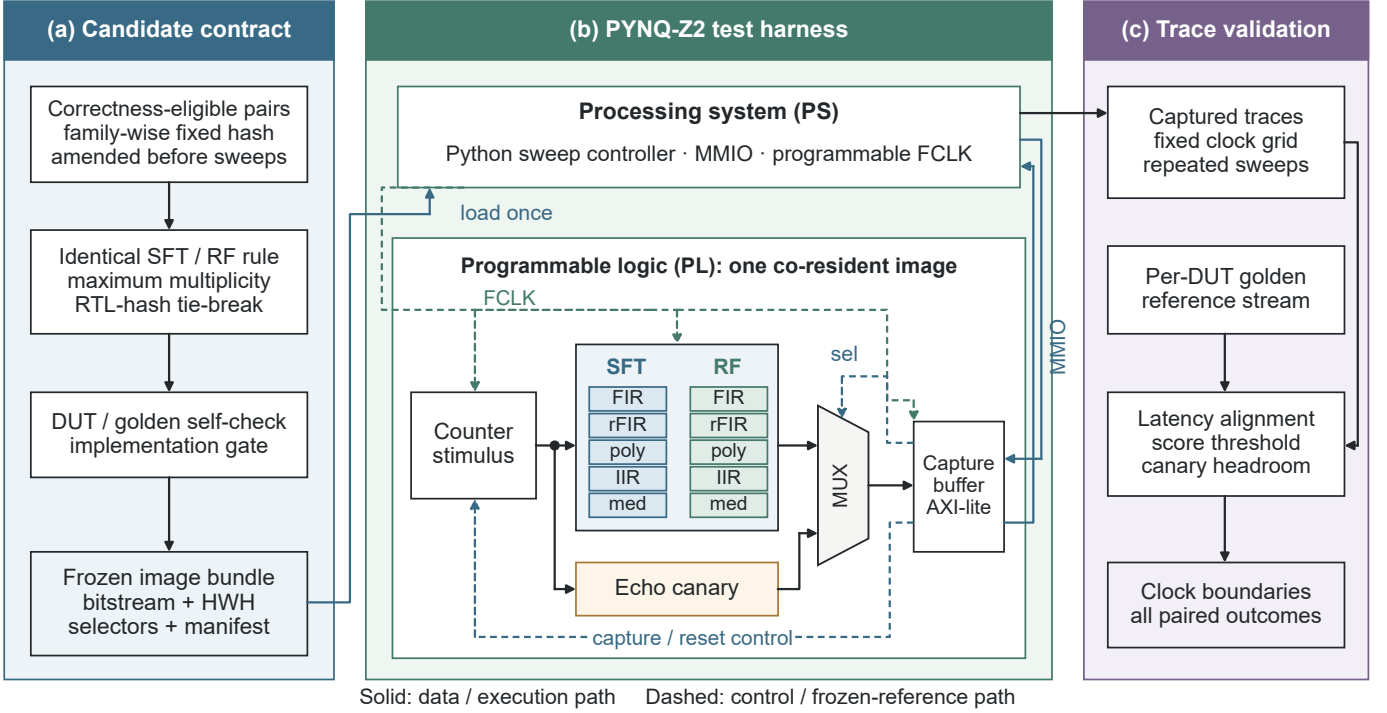


Fig. 2. Descriptive board-validation architecture. (a) The amended correctness-eligible selection uses the same multiplicity and hash rules for SFT and RF, with no frequency-based reselection. (b) The processing system configures the clock and accesses the capture buffer through memory-mapped AXI. Counter-driven SFT/RF DUTs and the echo canary coexist in one programmable-logic image; the selector chooses the captured output. Solid paths carry data or execution flow; dashed paths show clock and capture control. Logical buses are simplified for readability. (c) Captured traces are compared with per-DUT golden streams, and clock boundaries are accepted only with the specified canary headroom.

with respect to that split, but it is not an independent validation of the post-diagnosis gate scope.

All frozen inputs and outputs are content-hashed. Generated claim ledgers bind manuscript values to artifact line ranges; figure manifests additionally store the exact plotted arrays and input/output hashes. Failed protocol attempts, implementation failures, board amendments, and excluded exploratory results remain in the public record rather than being overwritten.

V. RESULTS

A. Sealed physical efficacy

Figure 3 shows every design and the frozen paired analysis. Across 20 held-out designs, the repaired-RF arm raises penalized equal-sample F_{max} from 35.4 to 93.0 MHz. The paired gain is 57.6 MHz with interval [37.7, 77.2] MHz.

The result is not confined to the easier stratum. Interpolation improves by 77.9 MHz with interval [48.4, 107.2], while extrapolation improves by 37.4 MHz with interval [13.9, 60.9] MHz. All zero-valued designs remain in Fig. 3; the aggregate is therefore not obtained by conditioning on successful hardware.

Both independently trained RF policies improve: the seed-specific overall effects are 49.5 and 65.8 MHz. Their positive directions also hold separately under interpolation and extrapolation (Table III). Correctness changes from 56.2% for SFT to 53.9% for RF. This difference is not hidden: because every incorrect draw scores zero, it is already paid for in the primary effect.

The post-primary family breakdown prevents the aggregate from becoming a universal family claim. FIR, reverse FIR, polynomial, and recursive-filter point estimates are positive, whereas median is negative and is represented only by extrapolation designs. These rows are descriptive because the family cells are small and unequal. The primary direction is nevertheless insensitive to two post-primary checks: the unstratified paired percentile bootstrap gives [31.1, 85.4] MHz, and the mean effect across leave-one-family-out omissions ranges from 46.0 to 69.7 MHz. Both checks are post-primary sensitivities, not replacements for the frozen interval.

B. Reward-package controls and endpoint decomposition

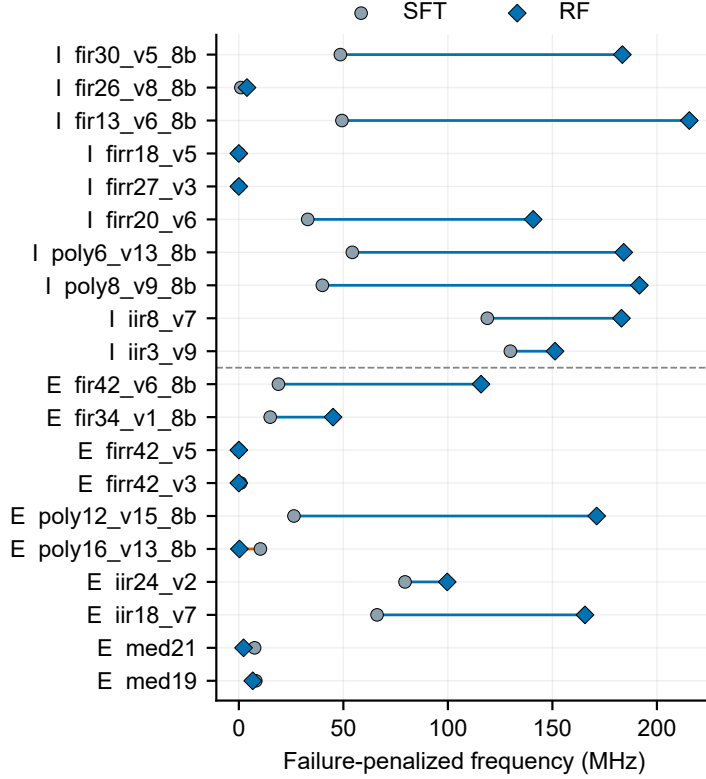
Table V separates functional yield from the physical quality of oracle-passing draws. With equal budgets across designs, pooled $q\mu$ reproduces the primary mean exactly; this pooled μ is not an unweighted mean of per-design conditional frequencies. RF increases pooled conditional frequency from 62.89 to 172.71 MHz while correctness decreases. In contrast, the correctness-only policy has a conditional frequency of 36.62 MHz. The observed RF gain is thus a shift in the physical quality of passing draws, not an increase in their count.

Figure 4 separates physical utility from functional yield. Correctness-only training reaches 67.4% correctness, above both SFT and RF, but attains only 24.7 MHz penalized F_{max} . Its paired SFT-relative effect is -10.7 MHz with interval [-14.3, -7.2] MHz. Increasing validity alone therefore does not

TABLE II
HEADLINE SEALED STUDY 2 COMPARISON OVER THE SAME 20 DESIGNS. PENALIZED FMAX IS AVERAGED EQUALLY OVER DESIGNS AND SCORES INCORRECT OR IMPLEMENTATION- FAILED SAMPLES AS ZERO. CONFIDENCE INTERVALS ARE PAIRED, STRATIFIED 95% BOOTSTRAP INTERVALS VERSUS THE SFT BASELINE.

Arm	Training seeds	Penalized Fmax [MHz]	Correct [%]	Δ vs. SFT [95% CI]
SFT	—	35.4	56.2	—
Correctness only	1	24.7	67.4	-10.7 [-14.3, -7.2]
Original MLP reward	2	66.1	49.9	30.8 [14.1, 46.7]
Repaired RF reward	2	93.0	53.9	57.6 [37.7, 77.2]

(a) Per-design physical endpoint



I: interpolation E: extrapolation

(b) Paired RF – SFT effects

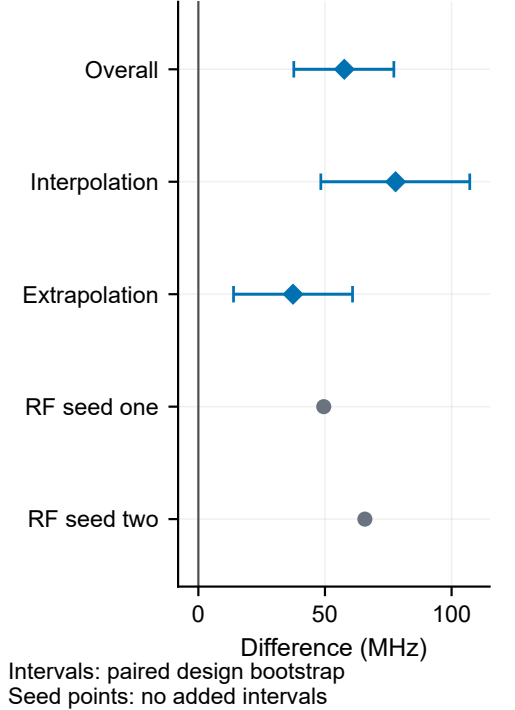


Fig. 3. Sealed primary efficacy. Panel (a) retains every design and joins its SFT and repaired-RF penalized equal-sample post-route F_{max} . Panel (b) reports the frozen paired effects and the separately displayed training-seed effects. The overall gain is 57.6 MHz [37.7, 77.2] MHz.

explain the hardware result in this realized seed. Its interval resamples designs conditional on that checkpoint; it does not establish a seed-robust ranking of correctness-only training.

The original MLP reward does improve over SFT, reaching 66.1 MHz with a paired gain of 30.8 MHz. The repaired RF nonetheless exceeds it directly by 26.9 MHz with interval [16.2, 37.5] MHz. The RF-minus-MLP intervals remain above zero under interpolation and extrapolation as well. Because the arms share their SFT initialization, tasks, update budget, oracle, and evaluation, this comparison favors the RF reward package for these checkpoints. It does not separate canonicalization, structural features, and predictor architecture, nor does it compare alternative policy-optimization algorithms.

C. Conditional physical-cost trade-off

The speed improvement is not a free PPA gain. On the 15-design common support, RF uses 754.9 rather than 689.2 LUTs and 236.2 rather than 161.4 flip-flops. Conversely, DSP use falls from 5.00 to 2.73 blocks, while BRAM remains unchanged. Vectorless power rises from 0.160 to 0.162 W. The paired intervals in Table VI show that the LUT, register, DSP, and vectorless-power directions are stable on common support. These conditional quantities answer what a compiled correct implementation costs; they do not replace the failure-penalized policy endpoint.

Table VII supplies a concrete CAD-level example of that aggregate trade-off. For fir13_v6_8b, the dominant SFT candidate stores delayed samples and products before a wide parallel sum. The dominant candidate from RF seed one instead emits a registered transposed multiply-accumulate chain. Vivado

TABLE III

INDEPENDENT REPAIRED-RF TRAINING-SEED EFFECTS ON THE SEALED ENDPOINT. SEED ROWS ARE SEPARATELY REPORTED POINT ESTIMATES; THE SEED-AVERAGED ROW USES THE FROZEN PAIRED, STRATIFIED 95% BOOTSTRAP INTERVALS.

Estimate	Interpolation	Extrapolation	Overall
RF seed 1	75.6	23.4	49.5
RF seed 2	80.1	51.4	65.8
Seed average [95% CI]	77.9 [48.4, 107.2]	37.4 [13.9, 60.9]	57.6 [37.7, 77.2]

TABLE IV

POST-PRIMARY DESCRIPTIVE FAMILY BREAKDOWN OF THE SEALED FAILURE-PENALIZED ENDPOINT. I/E GIVES INTERPOLATION/EXTRAPOLATION DESIGN COUNTS. ROWS ARE DESCRIPTIVE POINT ESTIMATES; THE SMALL AND UNEQUAL FAMILY CELLS DO NOT SUPPORT STABLE FAMILY-LEVEL CONFIDENCE INTERVALS.

Family	Designs	I/E	SFT [MHz]	RF [MHz]	RF-SFT [MHz]
FIR	5	3/2	26.6	112.8	86.2
Reverse FIR	5	3/2	6.8	28.2	21.4
Polynomial	4	2/2	32.7	136.8	104.1
Recursive filter	4	2/2	98.6	149.9	51.3
Median filter	2	0/2	7.8	4.5	-3.4

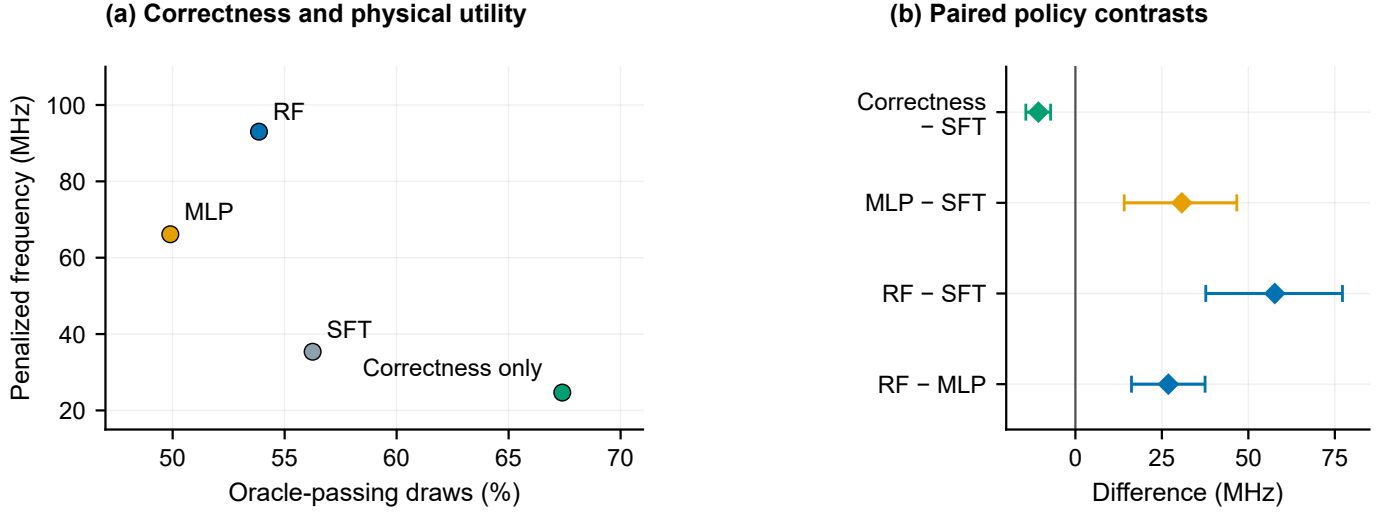


Fig. 4. Matched controls. Panel (a) plots correctness against the failure-penalized endpoint; the correctness-only arm is the key counterexample. Panel (b) gives paired contrasts. Repaired RF exceeds the original MLP reward by 26.9 MHz [16.2, 37.5] MHz.

TABLE V

POST-PRIMARY DECOMPOSITION ON THE COMPLETE SEALED DRAW SET.
CONDITIONAL FREQUENCY POOLS CORRECT DRAWS, INCLUDING ZERO-VALUED
IMPLEMENTATION FAILURES.

Policy	q (%)	μ (MHz)	$q\mu$ (MHz)
SFT	56.25	62.89	35.38
RF	53.85	172.71	93.01
MLP	49.90	132.56	66.14
Correctness only	67.40	36.62	24.68

maps the latter to substantially more LUTs and registers, no DSP blocks, and a much shorter timing path. Because this case was selected post-primary for interpretability, it illustrates a physically meaningful RTL transformation but does not establish that every family or every gain follows the same transformation.

D. Setup-closure sensitivity

The selected-candidate pilot yields Spearman correlation 0.963 between original timing-derived and setup-passing frequencies, with median symmetric percentage error 3.50%. None of the 3 non-tied pairs reverses direction (Table VIII). However, the largest candidate error is 45.56%: the reverse-FIR RF candidate changes from 211.19 to 335.81 MHz. The median error therefore must not be read as a uniform accuracy guarantee. The check supports relative ordering on this subset, while absolute frequency remains sensitive to implementation conditions. Setup passing alone is not full timing sign-off.

E. Descriptive PYNQ-Z2 evidence

Figure 5 reports all paired board outcomes under the architecture in Fig. 2. Across 3 complete sweeps, the maximum within-entry boundary spread is 0.0 MHz. The echo canary reaches 200.0 MHz and remains at least 57.1 MHz above every DUT boundary.

TABLE VI

CONDITIONAL PHYSICAL-COST DIAGNOSTICS ON THE 15-DESIGN THREE-ARM COMMON SUPPORT. VALUES ARE MULTIPLICITY-WEIGHTED WITHIN DESIGN AND EQUALLY WEIGHTED ACROSS DESIGNS. POWER IS VIVADO'S VECTORLESS ESTIMATE, NOT BOARD POWER.

Metric	SFT	MLP	RF	RF-SFT difference [CI]
LUTs	689.2	766.7	754.9	65.7 [47.2, 84.1]
Flip-flops	161.4	222.3	236.2	74.8 [66.1, 83.4]
DSP blocks	5.00	4.06	2.73	-2.26 [-2.82, -1.70]
BRAM blocks	0.00	0.00	0.00	0.00 [0.00, 0.00]
Vectorless power [W]	0.160	0.168	0.162	0.003 [0.002, 0.003]

TABLE VII

ILLUSTRATIVE SOURCE-LEVEL CONTRAST FOR fir13_v6_8b. THE ENTRIES ARE THE DOMINANT SFT CANDIDATE AND DOMINANT CANDIDATE FROM RF SEED ONE FOR THAT DESIGN. SELECTION OCCURRED AFTER MANUSCRIPT ASSESSMENT; THIS EXAMPLE EXPLAINS A POSSIBLE CAD MECHANISM BUT IS NOT A POPULATION OR CAUSAL ESTIMATE.

Arm	Observed RTL organization	Sample mass [%]	Fmax [MHz]	LUT	FF	DSP
SFT	Delayed samples and product registers followed by a parallel final sum	45.8	47.4	1	88	13
RF seed one	Registered transposed multiply-accumulate chain	87.5	220.1	251	221	0

TABLE VIII

DESCRIPTIVE SETUP-CLOSURE SENSITIVITY (MHz). ORIGINAL: FROZEN WNS-DERIVED VALUE. SETUP: CONSERVATIVE PASSING ENDPOINT UNDER THE PILOT FLOW.

Design	SFT orig.	SFT setup	RF orig.	RF setup
fir42_v6_8b	20.49	20.65	179.40	185.19
firr20_v6	40.92	40.29	211.19	335.81
iir18_v7	184.84	177.89	184.84	177.89
med21	10.96	10.47	10.96	10.47
poly6_v13_8b	34.64	34.64	192.12	192.12

TABLE IX

DESCRIPTIVE PYNQ-Z2 CLOCK-SWEEP CONFIRMATION ON 5 PAIRED DESIGNS. VALUES ARE MEDIANS OVER 3 REPEATED SWEEPS. THE SUBSET REQUIRED A DISCLOSED CORRECTNESS-FEASIBILITY AMENDMENT AND IS NOT INDEPENDENT CONFIRMATORY EVIDENCE.

Family	Design	SFT	RF	RF-SFT [MHz]
FIR	fir34_v1_8b	40.0	125.0	+85.0
FIRR	firr20_v6	58.8	142.9	+84.0
POLY	poly16_v13_8b	30.3	30.3	+0.0
IIR	iir18_v7	125.0	125.0	+0.0
MED	med19	40.0	30.3	-9.7

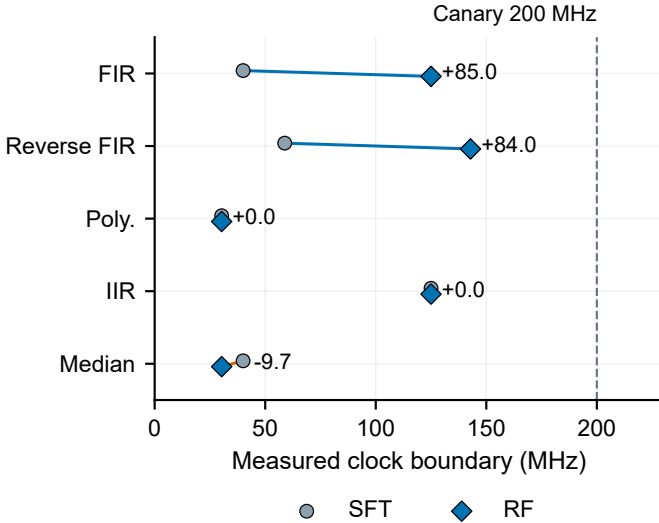


Fig. 5. Measured PYNQ-Z2 clock boundaries under the amended protocol. All pairs are retained: 2 RF wins, 2 ties, and 1 loss. Annotations are signed RF-minus-SFT differences in MHz; the dashed line is the echo-canary boundary. No population interval is inferred from this subset.

The descriptive mean changes from 58.8 to 90.7 MHz, a +31.9-MHz paired difference. That mean is driven by two large feed-forward gains. Polynomial and IIR tie, while the median pair changes by -9.7 MHz. The silicon evidence

therefore supports the main direction heterogeneously; it does not establish a universal family effect or independently confirm the population estimate.

F. Breadth and backbone transfer

The earlier, separate 30-design evaluation supplies broader five-family support. Its penalized endpoint rises from 73.0 to 176.7 MHz, with 27 of 30 designs improving. Feed-forward families show the most consistent gains; IIR is mixed and median remains slow in absolute terms. Because its uncertainty analysis is historical and post-hoc, this result supports breadth but is not pooled with the sealed study.

The full training recipe also transfers to Qwen-Coder on the earlier 22-design, 3-family universe. Penalized F_{max} rises from 64.9 to 200.5 MHz, a paired gain of 135.6 MHz with interval [122.7, 147.3] MHz; all 22 designs improve. Correctness also rises from 80.1% to 93.0%. This reduces a model-specific explanation, but its narrower family universe prevents direct pooling.

G. Probability reallocation and sampling efficiency

Repeating the support diagnostic on the sealed samples refines the mechanism claim. Among 16 designs with correct sampled strings under both policies, SFT's measured support

TABLE X
MATCHED POST-PRIMARY GENERAL-RTL EVALUATION. EACH CHECKPOINT USES
THE SAME BENCHMARK, GENERATION SETTINGS, AND DRAW BUDGET.

Checkpoint	Pass-at-one (%)
SFT	22.34
RF seed one	21.12
RF seed two	20.77

contains a candidate at least as fast as the RF conditional mean on 14. Mean inverse-Simpson effective correct-string support contracts from 3.78 to 2.67. Thus reallocation explains most common-support cases, but the sealed exceptions prevent the stronger claim that every optimized implementation was already observed in the finite SFT sample. Nor do unseen SFT samples permit a claim of open-ended invention.

On the sealed distribution, one RF draw lies between the empirical SFT perfect-selector expectations for 17 draws (92.47 MHz) and 18 draws (93.66 MHz). At the upper integer crossing, the SFT workload is 18 generations and 36 oracle simulation streams per design, with 10.125 expected correct-draw implementation attempts without caching or 3.078 distinct implementations with content caching. These are empirical expected workloads, not measured runtime or speedup. They exclude policy-training cost and assume perfect post-route selection on observed support; they do not imply dominance over an unrestricted SFT search.

H. Contextual bounds

The same-task API control shows why conditional speed is insufficient. A timing-focused prompt raises conditional F_{max} from 52.8 to 239.9 MHz while correctness changes from 27.1% to 22.9%; its penalized endpoint is 55.0 MHz. This is one API configuration with a smaller budget and is not a general frontier-model ranking.

For selected implementation quality, the fastest measured optimized RTL candidate reaches 0.96 $\times$ the geometric-mean throughput of engineer-pragma HLS over 19 comparable designs. It reaches 46.0 $\times$ no-pragma HLS. Because HLS starts from engineer-written C++ and the RTL value is selected rather than policy-average, this comparison is context, not the causal baseline.

I. Matched general-RTL specialization cost

The matched VerilogEval extension compares the actual sealed checkpoints on 156 official problems with 20 draws per checkpoint and problem. Pass-at-one is 22.34% for SFT, 21.12% for RF seed one, and 20.77% for RF seed two (Table X). The RF-seed-mean paired change is -1.39 percentage points with interval [-2.69, -0.10]. This indicates a small additional general-RTL regression, not “no further degradation.” The interval resamples problems conditional on the evaluated checkpoints and does not estimate policy-training seed variance.

The earlier base-model comparison showed pass-at-one decreasing from 32.2% to 16.0% after domain SFT. That experiment uses different generation settings; its absolute values are

not comparable with the matched RF extension and are not pooled. Together, the two experiments identify specialization costs at different training stages, without establishing performance on unseen industrial RTL.

J. Bounded reward-reliability diagnostic

Figure 6 contrasts the older exploited reward with the repaired RF trajectory. In the historical run, early penalized physical performance rises from 76.4 to 230.5 MHz. Late in training, however, the proxy rises from 216.6 to 497.9 MHz while correctness falls from 95.3% to 88.0%. Penalized physical performance therefore falls from 230.9 to 212.3 MHz.

A lexical audit identifies one plausible channel in the older reward: a line-anchored assignment feature increases while its layout-canonical counterpart does not, and diagnostic rescoring removes 101.2% of the raw late reward jump. The canonical scores are clamp-saturated and there are 0 token-identical cross-checkpoint pairs, so this is bounded attribution rather than a randomized causal test. In the repaired run, interpolation remains essentially stable from midpoint to endpoint and extrapolation improves. The positive repair claim comes from the sealed efficacy study; the historical trajectory only explains why the reporting boundary is necessary.

VI. DISCUSSION

A. Interpretation of the physical gain

The sealed study supports a distribution-level improvement within the studied accelerator generators: the RF-trained policies produce higher timing-derived physical utility per draw after failures are charged to the originating policy. The decomposition locates this gain in the physical quality of oracle-passing draws, not in higher correctness. This distinction matters when using a generator as an engineering tool, because a fast selected example and a reliably useful draw imply different verification and implementation costs.

The controls constrain, but do not fully identify, the cause. The original MLP reward improves on SFT, while RF improves further. RF changes the predictor class, canonicalization, and feature representation jointly; their individual contributions remain unresolved. Correctness-only training yields more passing samples but lower physical utility in its single primary seed. This is a counterexample to identifying correctness improvement with physical improvement, not evidence that correctness-only training must fail across seeds or models.

Candidate probabilities suggest a plausible mechanism. SFT often already samples a fast implementation, but infrequently; optimization increases mass on fewer correct outputs. The sealed support exceptions and finite draw budget prevent a claim that all optimized implementations were already available in the observed SFT sample. The illustrative FIR pair supplies a concrete direct-to-transposed structural contrast and a changed DSP/LUT mapping, but does not identify a universal topology change behind the aggregate effect.

B. Implications for CAD workflows

A learned reward is useful only insofar as it orders enough on-policy candidates to guide optimization. The measured

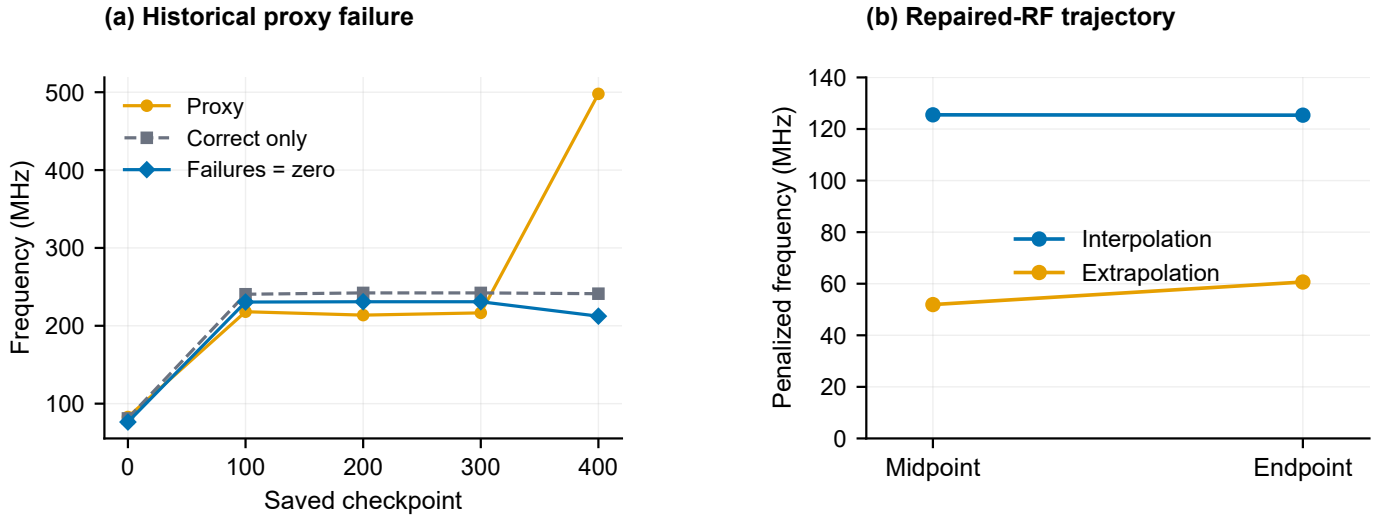


Fig. 6. Bounded reward-reliability evidence. Panel (a) shows a historical proxy rising while failure-penalized Vivado utility falls. Panel (b) shows the repaired RF midpoint-to-endpoint trajectory: interpolation changes from 125.5 to 125.4 MHz and extrapolation from 51.9 to 60.7 MHz.

reward resolution and sealed efficacy support that role for the repaired package; they do not validate every prediction. The historical late trajectory demonstrates why proxy progress cannot replace implementation evidence. Canonicalization reduces sensitivity to formatting and names, while compilation and sampled trace checks remain necessary to test the actual reward path.

The physical benefit comes with engineering costs. RF uses more LUTs and registers, fewer DSP blocks, and slightly more vectorless power on common correct support. Implementations may also differ in latency within the oracle’s alignment window: higher clock frequency is not automatically lower end-to-end latency. A deployed design would need explicit resource, latency, and power constraints, or a multi-objective reward, rather than a frequency-only preference. The matched VerilogEval regression further argues for reporting general competence alongside domain-specific physical gains.

The empirical perfect-selector crossing describes inference workloads on a finite support. It omits the cost of domain SFT, reward labels, and policy training. Matching non-flat optimizer updates also leaves attempted groups and compute unequal between arms. Consequently, the present evidence does not establish an end-to-end runtime advantage, energy saving, or deployment break-even point.

C. Threats to validity

a) Template-bounded generalization: The frozen split excludes evaluated designs from domain training, but tests new parameters within known generators rather than new specification families. It does not establish competence on control-dominated systems, memories, multiple clocks, or free-form industrial RTL. The median family occurs only under extrapolation, confounding family and regime in that cell. The separate Qwen-Coder result broadens backbone evidence but uses the earlier, narrower family universe. Neither split construction nor VerilogEval evaluation can exclude unknown overlap with a base model’s pretraining corpus.

b) Training variance and statistical resolution: The RF seeds share one SFT checkpoint, reward dataset, and benchmark generator; they are policy-training replications, not independent full-pipeline repeats. The correctness-only primary control has one seed. Design-bootstrap intervals condition on these realized checkpoints and the frozen task universe. Small, unequal family-by-regime cells limit their resolution. Post-primary unstratified and leave-one-family-out checks preserve the direction but do not remove these limitations. RF versus SFT is the sole prespecified primary contrast; the other intervals are secondary, without multiplicity adjustment.

c) Repair and validation history: The canonical-feature RF hypothesis was motivated by the historical MLP failure. Its efficacy test was frozen before the sealed split was opened, but the hypothesis was not selected blindly. The earlier all-nonempty-candidate validation contract also failed on an oracle-rejected completion. The revised reward-eligible contract reused unchanged checkpoints and an unseen efficacy split, but its population was chosen after diagnosing that failure. These facts limit how independently the repair and validation scope were selected.

d) Functional and physical validity: The oracle checks sampled, latency-aligned traces with a near-unity threshold, not exact equivalence over all inputs. Stronger exact-match and formal audits would be needed to elevate this correctness claim. The original physical endpoint comes from one device and a single requested-period flow. Its script does not fail closed when a valid setup path is absent: the slack variable has a zero default. The retained primary result has not been reclassified on the basis of that potential failure mode; a complete path-coverage audit is needed before asserting that every row satisfies a stronger timing contract.

The setup-closure pilot adds constraint-coverage checks and period bracketing on a small selected subset, under a newer Vivado version and constraints applied before synthesis. It supports relative ordering there, but the reverse-FIR outlier prevents interpreting the median error as universal calibra-

tion. The pilot does not enforce hold or pulse-width closure and therefore does not establish full timing sign-off. Cross-tool, cross-device, and operating-corner generalization remain untested.

e) Conditional costs, silicon, and baseline comparability:

Resource and power comparisons condition on common correct support, whereas the primary endpoint penalizes failures. The board subset was amended for correctness feasibility and sweep coverage before measurement, uses the same device family, and includes an RF loss. It is descriptive support, not an independent population confirmation. API prompting and selected-candidate HLS have different budgets and reporting objects. The study also lacks an executed matched DPO or recent external RTL-optimization baseline; internal reward controls do not establish superiority over those methods.

D. Artifact and claim traceability

The artifact package contains frozen manifests, sampled RTL and multiplicity, oracle summaries, implementation rows and failures, board traces, canonicalization audits, and generated claim ledgers.¹ Regeneration rejects missing design coverage, changed denominators, missing implementation rows, stale generated outputs, and unresolved numerical claims. The completed post-primary decomposition, setup-closure pilot, and matched VerilogEval archive are validated independently before their claims enter the manuscript.

Additional policy-training runs, prepared baseline protocols, and benchmark preflights are not reported as completed physical evaluations. The historical asymmetric board comparison and incomplete smaller-student evaluation remain excluded. New seed endpoints must be reported as a separate extension rather than silently changing the original sealed population or interval.

VII. CONCLUSION

Correctness-gated, proxy-guided policy optimization improves the physical utility of a fixed-budget RTL draw within the studied streaming-accelerator domain. On the sealed split, the RF reward package raises failure-penalized post-route timing-derived frequency from 35.4 to 93.0 MHz, with positive effects for both policy-training seeds. The gain reflects faster oracle-passing draws rather than higher correctness. Original-MLP and single-seed correctness-only controls constrain the explanation without isolating individual repair components.

The improvement is neither universal across families nor free of cost. Higher LUT/register use, slightly higher vectorless power, and a small matched VerilogEval regression accompany the physical gain. Selected setup-closure and board checks add directional evidence but do not replace the sealed endpoint or establish full timing sign-off. The resulting CAD lesson is specific: a learned proxy can guide a useful distribution shift, but functional testing, complete failure accounting, and independent physical evaluation must remain separate from the training objective.

¹Repository: <https://github.com/MoAmineHallam/FPGA>. The final release must identify the exact manuscript and artifact revision.

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